MILITARY SPECIFICATION

MICROCIRCUITS, LINEAR, PRECISION TIMERS,
MONOLITHIC SILICON

This amendment forms a part of MIL-M-38510/109A, dated 18 September 1980, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.2.1, under Device type, to the right of 01: Add "1/".

* 1.2.3, under letter P, add the following new letter and applicable information:

"2 C-2 (20-terminal, .350" x .350"), square chip carrier package"

1.4: Delete "1/" and substitute "2/".

Bottom of page, delete "1/" and substitute "2/". Add new note 1/:

" 1/ Warning: Device type 01 in the glass frit sealed C and P packages does not pass the requirement of glassivation layer integrity test, MIL-STD-883, test method 2021, as specified in MIL-M-38510. Therefore, the glassivation layer integrity test is waived for device type 01 in the glass frit sealed C and P packages only. Operation of the 01 device in these package types in the high level output voltage mode, may result in device reliability problems."

PAGE 2

* 1.5, "Maximum allowable power dissipation" column: Delete "T_C" (three places) and substitute "T_A" (three places).

* 1.5, under letter P, add the following new letter and applicable information:

"2	20 terminal square chip carrier package	330 mW at T_A = +125°C	+55°C/W	+121°C/W"

* 3.2.2, delete and substitute the following:

"3.2.2 Schematic circuit. The schematic circuit shall be submitted to the preparing activity prior to inclusion of manufacturer's device in this specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturer's schematics shall be maintained by the agent activity and will be available upon request. The schematic circuit shall be as specified on figure 2."

* 3.4, line 2: Delete "case" and substitute "ambient".

PAGES 3 THROUGH 9

* TABLE I: Delete all references to "T_C" and substitute "T_A".

* TABLE I, Trigger current test, Conditions column, delete and substitute the following:

" V_{CC} = 16.5 V dc for V_{TR} = 5.0 V dc
device types 01,03
V_{CC} = 15.0 V dc for V_{TR} = 4.5 V dc
device type 02 "

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* TABLE II, MIL-STD-883 test requirements column, Interim electrical parameters: Delete "(pre burn-in)".

* TABLE II, Class C devices column: Delete in its entirety.

* 3.6, second sentence: Delete in its entirety.

* 4.2b, delete and substitute: "b. Reverse bias burn-in (method 1015 of MIL-STD-883) - Reverse bias burn-in for class S devices is not required."

* 4.2d, second and third sentences, delete and substitute: "The percent defective allowable (PDA) shall be as specified in MIL-M-38510. The PDA shall be based on failures from group A, subgroup 1 test after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements.

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Add the following new figure:

Device types 01

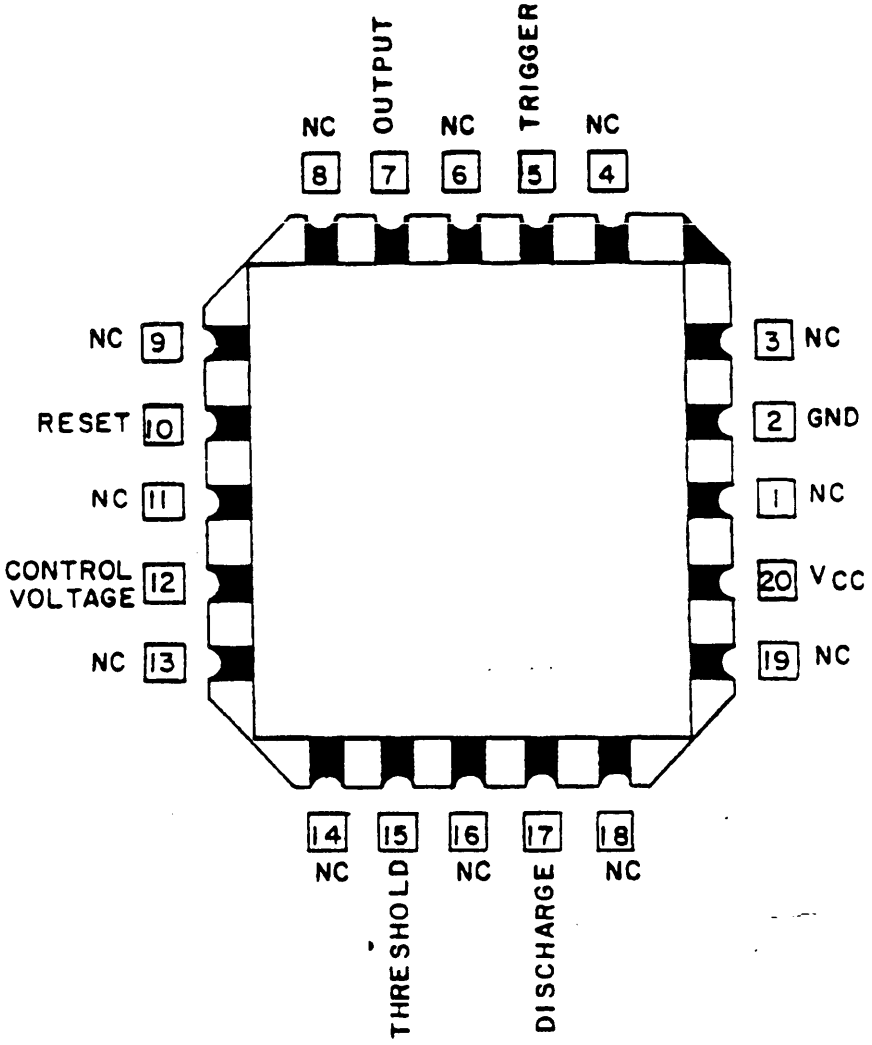
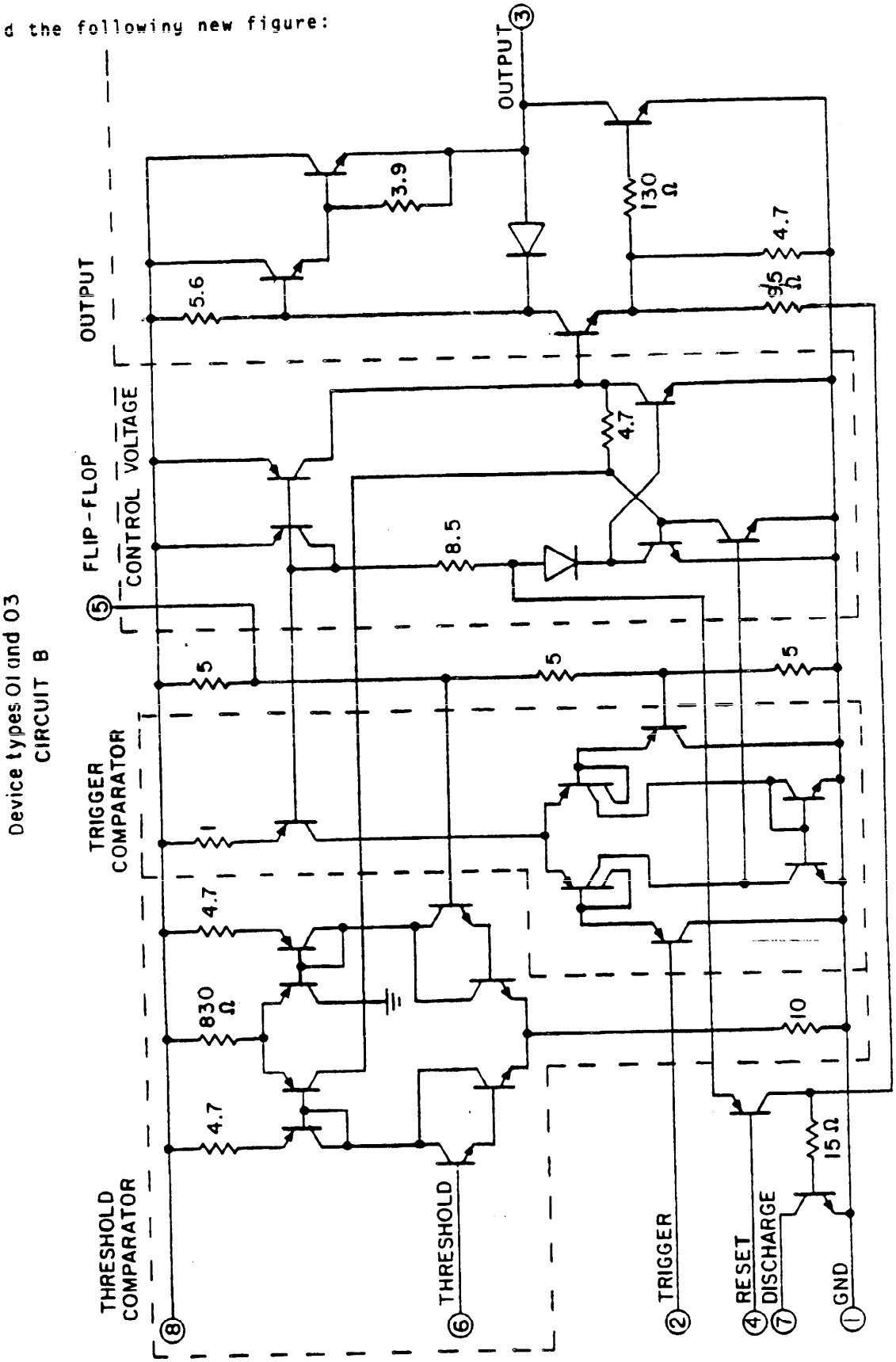


FIGURE 1. Terminal connections - Continued.

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Add the following new figure:



All resistance values are in kilohms unless otherwise specified.

FIGURE 2. Schematic Circuit - Continued.

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* Figure 6, add a normally closed relay (K4) between the out trigger pin and connection for K2.

PAGES 21 THROUGH 24

* TABLE III: Delete all references to "T_C" and substitute "T_A".

* TABLE III, tests 3,4,6,7,22,23,25,27, Pin measured, No. column: Delete "11" and substitute "7". Tests 5,8,24,27, Pin measured, No. column: Delete "11" and substitute "8". Tests 5,24, Relays closed column: Delete "2". Tests 8,27: Add "4".

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* TABLE III, tests 41,42,44,45, Pin measured, No. column: Delete "11" and substitute "7". Tests 43,46, Pin measured, No. column: Delete "11" and substitute "8". Test 43, Relays closed column: Delete "2". Test 46, Relays closed column: Add "4".

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4.4.3, add the following new subparagraph:

"c. Special subgroups shall be added to the group C inspection and shall consist of group A, subgroups 10 and 11 as specified in table III herein."

* 4.5.2, delete and substitute:

"4.5.2 Life test and burn-in cooldown procedure. When devices are measured at +25°C following application of the steady-state life or burn-in test condition, they shall be cooled to within +10°C of their power stable condition (at room temperature) prior to removal of the bias."

* TABLE IV, title, line 2: Delete "T_C = 25°C" and substitute "T_A = +25°C".

The margins of this amendment are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
NAVY - OS, SH
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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